Rockchip GMS测试 摄像头模块帮助文档

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前言

概述

本文提供了GMS测试过程中Camera模块常见fail项的处理方法。

产品版本

Android 14.0及以上版本

读者对象

本文档（本指南）主要适用于以下工程师：

技术支持工程师

软件开发工程师

修订记录

版本号	作者	修改日期	修改说明
V1.0.0	王潘祯撰	2025-04-18	初始版本
V1.1.0	王潘祯撰	2025-05-29	新增CTS/ITS测试失败案例描述
V1.2.0	王潘祯撰	2025-09-12	1. CTS-Verifier BYOD Managed Provisioning测试项 2. 更新无Camera情况，测试问题分析处理
V1.3.0	王潘祯撰	2025-11-06	新增Android16.0 常见失败项案例

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1. 概述

GMS测试包中包含大量摄像头相关（Camera）的模块，测试过程中会出现各种各样的fail，一些fail项需要找代码开发者去解决，一些fail项可以通过修改配置自行解决。本文提供了测试模块快速测试方法以及常见fail项解决办法。

2. 命令行测试

GMS测试正常流程是需要机器连接Ubuntu系统，并通过GMS测试包完成测试，测试时间相对比较久，在debug过程中如果通过正常流程去验证修改，效率偏低。

可以通过adb shell用命令行的方式通过am instrument启动Instrumentation测试。

以下以CtsCameraTestCases为例详细说明。

2.1 测试环境准备

2.1.1 CTS测试应用安装及权限设置

apk路径：

```
android-cts兼容性测试包/android-
cts/testcases/CtsCameraTestCases/arm64/CtsCameraTestCases.apk
```

apk安装：

```
adb install CtsCameraTestCases.apk
```

权限设置：

```
Settings--Apps--All apps--android.camera.cts
手动打开所需权限
```

2.2 测试命令

以CtsCameraTestCases中某一个case为例单测说明。

某一单项case：

```
android.hardware.camera2.cts.CaptureRequestTest#testEffectModes[1]
```

测试完整命令：

```
am instrument -w -r -e debug false -e class
android.hardware.camera2.cts.CaptureRequestTest#testEffectModes[1]
android.camera.cts/androidx.test.runner.AndroidJUnitRunner
```

测试结果：

```
130|rk3576_tablet:/ #
am instrument -w -r -e debug false -e class
android.hardware.camera2.cts.CaptureRequestTest#testEffectModes[1]
android.camera.cts/androidx.test.runner.AndroidJUnitRunner
INSTRUMENTATION_STATUS: class=android.hardware.camera2.cts.CaptureRequestTest
INSTRUMENTATION_STATUS: current=1
INSTRUMENTATION_STATUS: id=AndroidJUnitRunner
INSTRUMENTATION_STATUS: numtests=1
INSTRUMENTATION_STATUS: stream=
android.hardware.camera2.cts.CaptureRequestTest:
INSTRUMENTATION_STATUS: test=testEffectModes[1]
INSTRUMENTATION_STATUS_CODE: 1
INSTRUMENTATION_STATUS: class=android.hardware.camera2.cts.CaptureRequestTest
INSTRUMENTATION_STATUS: current=1
INSTRUMENTATION_STATUS: id=AndroidJUnitRunner
INSTRUMENTATION_STATUS: numtests=1
INSTRUMENTATION_STATUS: stream=.
INSTRUMENTATION_STATUS: test=testEffectModes[1]
INSTRUMENTATION_STATUS_CODE: 0
INSTRUMENTATION_RESULT: stream=

Time: 5.702

OK (1 test)

INSTRUMENTATION_CODE: -1
```

2.2.1 命令行说明

2.2.1.1 命令行格式

```
am instrument -w -r -e class testcase test_package/runner_class
```

2.2.1.2 参数说明

```
-w:保持adb shell打开直至测试完成
-r:以原始形式输出测试结果
-e <NAME> <VALUE>: set argument <NAME> to <VALUE>.提供过滤器和参数
```

2.2.1.3 获取test_package/runner_class

通过pm list instrumentation找test_package/runner_class

```
rk3576_tablet:/ #  pm list instrumentation
instrumentation:android.camera.cts/androidx.test.runner.AndroidJUnitRunner
(target=android.camera.cts)
```

CtsCameraTestCases的test_package/runner_class是：
android.camera.cts/androidx.test.runner.AndroidJUnitRunner

2.2.1.4 testcase

- 整个模块

  testcase：CtsCameraTestCases

- 模块中的一个大类集合

  testcase：android.hardware.camera2.cts.CaptureRequestTest

  测试命令：

```
am instrument -w -r -e debug false -e class
android.hardware.camera2.cts.CaptureRequestTest
android.camera.cts/androidx.test.runner.AndroidJUnitRunner
```

- 模块中的一个大类集合中的某一个case

  testcase：android.hardware.camera2.cts.CaptureRequestTest#testEffectModes[1]

  测试命令：

```
am instrument -w -r -e debug false -e class
android.hardware.camera2.cts.CaptureRequestTest#testEffectModes[1]
android.camera.cts/androidx.test.runner.AndroidJUnitRunner
```

3. Camera feature

3.1 Camera feature 配置情况

```
rk3576_tablet:/ # pm list features | grep cam
feature:android.hardware.camera  //后置Camera
feature:android.hardware.camera.any //包含camera feature默认会有
feature:android.hardware.camera.external //对应USB Camera
feature:android.hardware.camera.front //前置Camera
对应文件：
rk3576_tablet:/ # ls vendor/etc/permissions/android.hardware.camera.*
vendor/etc/permissions/android.hardware.camera.external.xml
vendor/etc/permissions/android.hardware.camera.front.xml
vendor/etc/permissions/android.hardware.camera.xml
```

3.2 Camera feature 选配位置

`sdk/device/rockchip/rk35xx`目录的 BoardConfig.mk配置支持的camera类型

```
# camera enable
BOARD_CAMERA_SUPPORT := true //配置是否支持MIPI Camera
BOARD_CAMERA_SUPPORT_EXT := true  //配置是否支持USB Camera
```

`sdk/device/rockchip/common/modules$` 中 `camera_aidl.mk`将对应的feature文件编译进板子

```
ifeq ($(BOARD_CAMERA_SUPPORT_EXT),true)
PRODUCT_COPY_FILES += \
frameworks/native/data/etc/android.hardware.camera.external.xml:$(TARGET_COPY_OUT
_VENDOR)/etc/permissions/android.hardware.camera.external.xml
endif

ifeq ($(BOARD_CAMERA_SUPPORT),true)
PRODUCT_COPY_FILES += \
frameworks/native/data/etc/android.hardware.camera.xml:$(TARGET_COPY_OUT_VENDOR)/
etc/permissions/android.hardware.camera.xml \
frameworks/native/data/etc/android.hardware.camera.front.xml:$(TARGET_COPY_OUT_VE
NDOR)/etc/permissions/android.hardware.camera.front.xml
endif
```

4. Android SDK版本

Android发布的不同SDK版本，提交点差异，可能测试失败项都不一样；所以提交redmine时候，建议同步提供板子目前使用的sdk版本情况。

获取SDK版本命令：`getprop | grep rksdk`

示例如下：

```
rk3576_tablet:/ # getprop | grep rksdk
[ro.rksdk.version]: [ANDROID14_RKR6]
```

5. CTS

5.1 Android 14.0 USB Camera CTS 常见失败项处理办法

5.1.1 帧率相关

5.1.1.1 测试项

```
android.hardware.camera2.cts.CaptureRequestTest#testNoiseReductionModeControl[1]
android.hardware.cts.CameraGLTest#testCameraToSurfaceTextureMetadata
android.hardware.cts.CameraTest#testPreviewFpsRange
```

5.1.1.2 报错日志

- android.hardware.camera2.cts.CaptureRequestTest#testNoiseReductionModeControl[1]

```
java.lang.Exception: Test failed for camera 111: Frame duration must be in the
range of [200000000, 500000000], value 110679333 is out of range [170000000,
575000000]
    at
android.hardware.camera2.cts.testcases.Camera2SurfaceViewTestCase.tearDown(Camera
2SurfaceViewTestCase.java:176)
    at
android.hardware.camera2.cts.CaptureRequestTest.tearDown(CaptureRequestTest.java:
152)
```

- android.hardware.cts.CameraGLTest#testCameraToSurfaceTextureMetadata

```
java.lang.AssertionError: Too many frame intervals out of frame rate bounds: 98,
limit 10
    at org.junit.Assert.fail(Assert.java:89)
    at org.junit.Assert.assertTrue(Assert.java:42)
    at android.hardware.cts.CameraGLTest$6.run(CameraGLTest.java:665)
    at android.hardware.cts.CameraGLTest.runForAllCameras(CameraGLTest.java:346)
    at
android.hardware.cts.CameraGLTest.testCameraToSurfaceTextureMetadata(CameraGLTest
.java:532)
```

- android.hardware.cts.CameraTest#testPreviewFpsRange

```
Test failed to run to completion. Reason: 'Test run failed to complete. Expected
777 tests, received 755'. Check device logcat for details
```

5.1.1.3 解决办法

`sdkhardware/rockchip/camera_aidl$` 目录添加如下修改，如没有更新SDK或找FAE获取

```
commit 21be8bc6487b27800c5a25612980fc400d55e9f3
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Fri Aug 23 10:21:08 2024 +0800

    device: usb camera: default use fpsranges [15:30] fix cts issue

    Test: CtsCameraTestCases

 android.hardware.camera2.cts.CaptureRequestTest#testNoiseReductionModeControl[1]
```

```
    failed log:
    Frame duration must be in the range of [500000000, 1000000000],
    value 64565200 is out of range [425000000, 1150000000])

    Type: Fix
    Redmine ID: #501921
    Associated modifications:
    https://10.10.10.29/c/android/platform/hardware/interfaces/+/181892
    Test: run camera cts

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: I33dca53101350f7680849ae69b0bb603de211b47
```

5.1.2 Camera feature相关

5.1.2.1 测试项

```
android.hardware.camera2.cts.CameraManagerTest#testCameraManagerGetDeviceIdList[1
]
```

5.1.2.2 报错日志

```
junit.framework.AssertionFailedError: External camera is not connected on device
with FEATURE_CAMERA_EXTERNAL
    at junit.framework.Assert.fail(Assert.java:50)
    at junit.framework.Assert.assertTrue(Assert.java:20)
    at
android.hardware.camera2.cts.CameraManagerTest.testCameraManagerGetDeviceIdList(C
ameraManagerTest.java:204)
```

5.1.2.3 解决办法

沟通分析客户反馈没有接USB Camera，但实际跑进去了，说明Camera相关feature中配置支持了USB Camera。
直接先将板上feature文件先删掉试验。

```
rk3576_tablet:/# rm vendor/etc/permissions/android.hardware.camera.external.xml
rk3576_tablet:/# reboot
```

测试pass后，参考修改如下

```
sdk/device/rockchip/rk35xx$ git diff
diff --git a/BoardConfig.mk b/BoardConfig.mk
index 610f2c1..640ecb2 100644
--- a/BoardConfig.mk
+++ b/BoardConfig.mk
@@ -103,9 +103,9 @@ BOARD_RKNN_SUPPORT := false
 BOARD_WIDEVINE_OEMCRYPTO_LEVEL := 3
```

```
 # camera enable
-BOARD_CAMERA_SUPPORT_EXT := true
+BOARD_CAMERA_SUPPORT_EXT := false
 CAMERA_SUPPORT_HDMI ?= false
 #CAMERA_SUPPORT_OSD := true
 #CAMERA_SUPPORT_VIRTUAL := true
```

5.1.3 时间戳相关

5.1.3.1 测试项

```
android.hardware.camera2.cts.ReadoutTimestampTest#testReadoutTimestamp
```

5.1.3.2 报错日志

```
java.lang.Exception: There were 40 errors:
  java.lang.Throwable(Test failed for camera 125: Camera 125 timestampBase 0
readoutTimestamp (0) should be equal to image timestamp (82754540506000 (expected
= 82754540506000, actual = 0) )
  java.lang.Throwable(Test failed for camera 125: Camera 125 timestampBase 0:
readoutTime (0) - captureStart time (82754540506000) should be > 0)
```

5.1.3.3 解决办法

`sdkhardware/rockchip/camera_aidl$` 目录添加如下修改，如没有更新SDK或找FAE获取

```
commit e57d39c542a7911364ea60ed70cada7cb77a4038
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Fri Aug 23 10:43:40 2024 +0800

    device: usb camera: add readoutTimestamp to fix cts issue

    Test: CtsCameraTestCases

 android.hardware.camera2.cts.ReadoutTimestampTest#testReadoutTimestamp

    failed log:
    readoutTimestamp (0) should be equal to image timestamp
    (82754540506000 (expected = 82754540506000, actual = 0) )

    Type: Fix
    Redmine ID: #501921
    Associated modifications:N/A
    Test: run camera cts

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: Id6450a0ee7907b9f0940e1c10083ee3500f8912
```

5.1.4 方向相关

5.1.4.1 测试项

```
android.hardware.camera2.cts.ExtendedCameraCharacteristicsTest#testCameraOrientationAlignedWithDevice[1]
```

5.1.4.2 报错日志

```
unit.framework.AssertionFailedError: Camera 100's long dimension must align with screen's long dimension expected:&lt;false&gt; but was:&lt;true&gt;
    at junit.framework.Assert.fail(Assert.java:50)
    at junit.framework.Assert.failNotEquals(Assert.java:287)
    at junit.framework.Assert.assertEquals(Assert.java:67)
    at junit.framework.Assert.assertEquals(Assert.java:147)
    at
android.hardware.camera2.cts.ExtendedCameraCharacteristicsTest.testCameraOrientationAlignedWithDevice(ExtendedCameraCharacteristicsTest.java:3309
```

5.1.4.3 解决办法

修改板上`/vendor/etc/external_camera_config.xml`中如下关键字段值

```
<!-- orientation -->
<Orientation  degree="90"/> <!-- 0/90/180/270 -->
```

在SDK上的位置：`sdk/device/rockchip/common/external_camera_config.xml`

5.1.5 Android14.0-R5-USB Camera补丁

修复Android14.0 R5版本，USB Camera CTS 通用失败项提交点如下：如果没有找FAE获取

```
提交点1：
commit 8847d6aa76c9445db5094c332dd7d99a7e1240e4
Author: Jian Qiu <qiujian@rock-chips.com>
Date:   Wed Jul 31 15:06:07 2024 +0800

    ExternalCamera: Dealing with V4L2_BUF_FLAG_ERROR

    Try DQBUF again when obtaining V4L2_BUF_FLAG_ERROR

    Type: Fix
    Redmine ID: #497268
    Associated modifications: N/A
    Test: N/A

    Signed-off-by: Jian Qiu <qiujian@rock-chips.com>
    Change-Id: I61d0c0aa69427984b8b2b65841bb9204a57ac83
```

提交点2：

```
commit d22ac4f020a7499e50ef8e120d77cdcdff837f6a
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Wed Aug 7 20:32:46 2024 +0800

    device: fix cts stream usage issue

    Test: CtsCameraTestCases
    android.hardware.camera2.cts.SurfaceViewPreviewTest#testSurfaceSet[1]

    Type: Fix
    Redmine ID: #500232
    Associated modifications: N/A
    Test: camera cts

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: Iba303389d5c0d25a6384a421a9b3a20cf198b045
```

提交点3：

```
commit 5398546acbb1b814ba0845c8fc7a04c478da6b7a
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Mon Aug 12 20:23:40 2024 +0800

    device: fix mipi camera cts readoutTimestamp issue

    Test: CtsCameraTestCases
    android.hardware.camera2.cts.ReadoutTimestampTest#testReadoutTimestamp

    Type: Fix/Function
    Redmine ID: #500232
    Associated modifications: gerrit links
    Test: camera cts

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: I0f3fef0cb576f37fe004006b844adb349987c4df
```

提交点4：

```
commit 5186fde7ebc54597b12e98520914597ec18cb7ca
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Tue Aug 27 10:26:38 2024 +0800

    dev: usb camera: set kBadFramesAfterStreamOn to 4 to fix cts issue

    testAllocationFromCameraFlexibleYuv test zero value of frame,
    for most of usb camera need skip 2-3 frame to get normal frame;
    so use default 4 frame to suite most cameras.

    Test: CtsCameraTestCases

android.hardware.camera2.cts.AllocationTest#testAllocationFromCameraFlexibleYuv[1
]

    failed log:

    junit.framework.AssertionFailedError: Input allocation data was not
    updated
        at junit.framework.Assert.fail(Assert.java:50)
        at junit.framework.Assert.assertTrue(Assert.java:20)
```

```
        at

android.hardware.camera2.cts.helpers.AssertHelpers.assertArrayNotAllZeroes(Assert
Helpers.java:48)
        at

android.hardware.camera2.cts.AllocationTest.validateInputOutputNotZeroes(Allocati
onTest.java:441)
        at

android.hardware.camera2.cts.AllocationTest.-$$Nest$mvalidateInputOutputNotZeroes
(UnknownSource:0)
        at

android.hardware.camera2.cts.AllocationTest$2$1$1.run(AllocationTest.java:511)

    Type: Fix
    Redmine ID: #501921
    Associated modifications: N/A
    Test: camera cts

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: Ic93d2e4670bfae601aea525b9fbd841ffa442453

提交点5：
commit 666d5aa4fa154c7454e3d61aa0d09c38a76ad12e
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Fri Aug 23 10:21:08 2024 +0800

    device: usb camera: default use fpsranges [15:30] fix cts issue

    Test: CtsCameraTestCases

android.hardware.camera2.cts.CaptureRequestTest#testNoiseReductionModeControl[1]

    failed log:
    Frame duration must be in the range of [500000000, 1000000000],
    value 64565200 is out of range [425000000, 1150000000])

    Type: Fix
    Redmine ID: #501921
    Associated modifications:
    https://10.10.10.29/c/android/platform/hardware/interfaces/+/181892
    Test: run camera cts

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: I33dca53101350f7680849ae69b0bb603de211b47

提交点6：
commit e0274c8ed40cce941f13d90f811fa013818c2f1f
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Fri Aug 23 10:43:40 2024 +0800

    device: usb camera: add readoutTimestamp to fix cts issue

    Test: CtsCameraTestCases
    android.hardware.camera2.cts.ReadoutTimestampTest#testReadoutTimestamp
```

```
    failed log:
    readoutTimestamp (0) should be equal to image timestamp
    (82754540506000 (expected = 82754540506000, actual = 0) )

    Type: Fix
    Redmine ID: #501921
    Associated modifications:N/A
    Test: run camera cts

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: Id6450a0ee7907b9f0940e1c10083ee3500f8912e
```

提交点7：

```
commit 34feb5b65d05a3a8dd860e292d3af17504086dec (tag: android-14.0-mid-rkr6)
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Thu Nov 14 16:07:56 2024 +0800

    device: ExternalCamera: fix usb camera video recording stuck issue

    processOneCaptureRequest and FrameWorkerThread::threadLoop not in
    the same thread, this cause when switch FPS, processOneCaptureRequest
    do Streamoff, but FrameWorkerThread::threadLoop is still do dqbuf, cause
    stuck issue

    Log:

    D ExtCamDevSsn: configureV4l2StreamLocked: V4L configuration
    format:MJPG, w 1280, h 720
    D ExtCamDevSsn: @v4l2StreamOffLocked: enter!
    D ExtCamDevSsn: close mBufFd[0]=22
    E ExtCamDevSsn: dequeueV4l2FrameLocked: DQBUF fails: Invalid argument
    E ExtCamDevSsn: threadLoop: V4L2 deque frame failed!
    D ExtCamDevSsn: close mBufFd[1]=23
    D ExtCamDevSsn: close mBufFd[2]=24
    E ExtCamDevSsn: dequeueV4l2FrameLocked: DQBUF fails: Invalid argument

    Type: Fix
    Redmine ID: # 518321
    Associated modifications:
    https://10.10.10.29/c/android/rk/hardware/rk35/camera_aidl/+/221884
    Test: open camera do video recording.
:
    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: I6a7ba57113202879f8bd2298c24c87073d5de625
```

打补丁位置：

`sdk/hardware/rockchip/camera_aidl/device$  mm -j12`

生成/vendor/lib64/camera.device-external-impl-rk.so库，替换板上库后，
使用如下命令重启服务生效：

```
adb shell "kill -9 `pidof android.hardware.camera.provider-V1-external-service-rk`"
```

5.2 Android 14.0 MIPI Camera CTS 常见失败项处理办法

5.2.1 camera3_profiles.xml配置错误导致失败项

5.2.1.1 aeAvailableModes配置相关

- 测试项

```
android.hardware.camera2.cts.CaptureRequestTest#testAeModeAndLock[1]
android.hardware.camera2.cts.CaptureRequestTest#testFlashControl[1]
android.hardware.camera2.cts.RobustnessTest#testSimultaneousTriggers[1]
android.hardware.camera2.cts.RobustnessTest#testAeAndAfCausality[1]
android.hardware.camera2.cts.RobustnessTest#testAfThenAeTrigger[1]
android.hardware.camera2.cts.RobustnessTest#testAeThenAfTrigger[1]
android.hardware.camera2.cts.RobustnessTest#testBasicTriggerSequence[1]
```

- 报错日志

```
junit.framework.AssertionFailedError: The static info key
'android.control.aeAvailableModes' values [0] must contain ON mode
    at junit.framework.Assert.fail(Assert.java:50)
    at
android.hardware.camera2.cts.helpers.StaticMetadata.failKeyCheck(StaticMetadata.j
ava:3051)
    at
android.hardware.camera2.cts.helpers.StaticMetadata.checkTrueForKey(StaticMetadat
a.java:3022)
    at
android.hardware.camera2.cts.helpers.StaticMetadata.getAeAvailableModesChecked(St
aticMetadata.java:1326)
    at
android.hardware.camera2.cts.CaptureRequestTest.testAeModeAndLock(CaptureRequestT
est.java:473)
```

- 解决办法

这类报错修改camera3_profiles.xml中关键字,

`<control.aeAvailableModes value="OFF"/>`

改成：

`<control.aeAvailableModes value="ON,OFF"/>`

5.2.1.2 不带马达摄像头模组测试

- 测试项

```
android.hardware.camera2.cts.CaptureRequestTest#testAfModes[1]
android.hardware.camera2.cts.StillCaptureTest#testFocalLengths[1]
```

- 报错日志

android.hardware.camera2.cts.CaptureRequestTest#testAfModes[1]

```
com.android.ex.camera2.exceptions.TimeoutRuntimeException: Unable to get the
expected result value [2, 6] for key android.control.afState after waiting for
100 results
    at
android.hardware.camera2.cts.CameraTestUtils.waitForAnyResultValue(CameraTestUtil
s.java:4000)
    at
android.hardware.camera2.cts.testcases.Camera2SurfaceViewTestCase.waitForAnyResul
tValue(Camera2SurfaceViewTestCase.java:371)
    at
android.hardware.camera2.cts.CaptureRequestTest.afModeTestByCamera(CaptureRequest
Test.java:2633)
    at
android.hardware.camera2.cts.CaptureRequestTest.testAfModes(CaptureRequestTest.ja
va:787)
```

android.hardware.camera2.cts.StillCaptureTest#testFocalLengths[1]

```
com.android.ex.camera2.exceptions.TimeoutRuntimeException: Unable to get the
expected result value [0] for key android.lens.state after waiting for 100
results
    at
android.hardware.camera2.cts.CameraTestUtils.waitForAnyResultValue(CameraTestUtil
s.java:4000)
    at
android.hardware.camera2.cts.CameraTestUtils.waitForResultValue(CameraTestUtils.j
ava:4023)
    at
android.hardware.camera2.cts.testcases.Camera2SurfaceViewTestCase.waitForResultVa
lue(Camera2SurfaceViewTestCase.java:348)
    at
android.hardware.camera2.cts.StillCaptureTest.focalLengthTestByCamera(StillCaptur
eTest.java:759)
    at
android.hardware.camera2.cts.StillCaptureTest.testFocalLengths(StillCaptureTest.j
ava:716)
```

- 解决办法

如果不支持马达，camera3_profiles.xml中马达相关配置如下：

```
<control.afAvailableModes value="OFF "/>
<control.maxRegions value="1,0,0"/>
<lens.info.minimumFocusDistance value="0.0"/> <!-- HAL may override this value
from CMC for RAW sensors -->
<request.availableRequestKeys中删除：control.afRegions 字段
<request.availableResultKeys中删除：  control.afRegions 字段
```

5.2.1.3 必配参数漏配失败项

- 测试项

```
android.hardware.camera2.cts.CameraDeviceTest#testCameraDeviceVideoSnapShotTemplate[1]
android.hardware.camera2.cts.CameraDeviceTest#testCameraDeviceRecordingTemplate[1]
android.hardware.camera2.cts.CameraDeviceTest#testCameraDevicePreviewTemplate[1]
android.hardware.camera2.cts.CameraDeviceTest#testCameraDeviceStillTemplate[1]
android.hardware.camera2.cts.CameraDeviceTest#testCameraDeviceZSLTemplate[1]
android.hardware.camera2.cts.StaticMetadataTest#testCapabilities[1]
```

- 报错日志

```
java.lang.Exception: Test failed for camera 0: Key android.control.afTrigger shouldn't be null
    at android.hardware.camera2.cts.testcases.Camera2AndroidTestCase.tearDown(Camera2AndroidTestCase.java:169)
    at android.hardware.camera2.cts.CameraDeviceTest.tearDown(CameraDeviceTest.java:158)
```

- 解决办法

camera3_profiles.xml中`request.availableResultKeys`和`request.availableResultKeyss`字段，

添加关键字`control.afTrigger`

5.2.2 拍照帧率相关

5.2.2.1 测试项

```
android.hardware.camera2.cts.CaptureRequestTest#testEdgeModeControl[1]
android.hardware.camera2.cts.CaptureRequestTest#testNoiseReductionModeControl[1]
```

5.2.2.2 报错日志

```
java.lang.Exception: There were 2 errors:
  java.lang.Throwable(Test failed for camera 0: Frame duration must be in the range of [33333333, 33333333], value 34311688 is out of range [32833332, 33833332])
  java.lang.Throwable(Test failed for camera 0: Frame duration must be in the range of [33333333, 33333333], value 34334488 is out of range [32833332, 33833332])
    at android.hardware.camera2.cts.testcases.Camera2SurfaceViewTestCase.tearDown(Camera2SurfaceViewTestCase.java:176)
    at android.hardware.camera2.cts.CaptureRequestTest.tearDown(CaptureRequestTest.java:152)
```

5.2.2.3 解决办法

`sdkhardware/rockchip/camera$` 目录参考如下修改，

`camera3_profiles.xml` 中添加关键字段：`sensor.frameDuration`

```
commit c8b282f057abb23763f1d8187016e56d3bd0a940
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Sat Nov 2 15:48:51 2019 +0800

    camera3_profiles: fix gc2145 following cts failed

    android.hardware.camera2.cts.CaptureRequestTest#testEdgeModeControl

android.hardware.camera2.cts.CaptureRequestTest#testNoiseReductionModeControl

    Logs:
    Throwable(Test failed for camera 0: Frame duration must be in the range of
[50000000, 50000000],
     value 61273000 is out of range [49250000, 50750000])

    Change-Id: Ib2ade725e9dbe4d8027998e1017a30c6322b7c28
    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
```

5.2.3 录像帧率相关

5.2.3.1 测试项

```
android.hardware.camera2.cts.RecordingTest#testBurstVideoSnapshot[1]
android.hardware.camera2.cts.RecordingTest#testIntermediateSurfaceRecording[1]
```

5.2.3.2 报错日志

```
E TestRunner: failed: testBurstVideoSnapshot[1]
(android.hardware.camera2.cts.RecordingTest)
E TestRunner: ----- begin exception -----
E TestRunner: junit.framework.AssertionFailedError: Camera 0: Video frame drop
rate too high: 97.142860%, tolerance 10.000000%. Video size: 1280x720,
              expectedDuration [3003.000000,3003.000000], expectedFrameDuration
33.333332, frameDropCnt 34, frameCount 35
E TestRunner:   at junit.framework.Assert.fail(Assert.java:50)
E TestRunner:   at junit.framework.Assert.assertTrue(Assert.java:20)
E TestRunner:   at
android.hardware.camera2.cts.RecordingTest.validateRecording(RecordingTest.java:2
587)
E TestRunner:   at
android.hardware.camera2.cts.RecordingTest.validateRecording(RecordingTest.java:2
489)
E TestRunner:   at
android.hardware.camera2.cts.RecordingTest.videoSnapshotTestByCamera(RecordingTes
t.java:2062)
```

```
E TestRunner:   at
android.hardware.camera2.cts.RecordingTest.videoSnapshotHelper(RecordingTest.java
:1807)
E TestRunner:   at
android.hardware.camera2.cts.RecordingTest.testBurstVideoSnapshot(RecordingTest.j
ava:696)
E TestRunner:   at java.lang.reflect.Method.invoke(Native Method)
E TestRunner:   at
org.junit.runners.model.FrameworkMethod$1.runReflectiveCall(FrameworkMethod.java:
59)

E TestRunner: failed: testIntermediateSurfaceRecording[1]
(android.hardware.camera2.cts.RecordingTest)
W libc    : Access denied finding property "sys.hwc.display_pipeline_timeline"
E TestRunner: ----- begin exception -----
E TestRunner: junit.framework.AssertionFailedError: Camera 0: Video duration
doesn't match: recorded 2416.000000ms, expected [1173.333374,1760.000000]ms.
E TestRunner:   at junit.framework.Assert.fail(Assert.java:50)
E TestRunner:   at junit.framework.Assert.assertTrue(Assert.java:20)
E TestRunner:   at
android.hardware.camera2.cts.RecordingTest.validateRecording(RecordingTest.java:2
552)
E TestRunner:   at
android.hardware.camera2.cts.RecordingTest.validateRecording(RecordingTest.java:2
489)
E TestRunner:   at
android.hardware.camera2.cts.RecordingTest.completeBasicRecording(RecordingTest.j
ava:1681)
E TestRunner:   at
android.hardware.camera2.cts.RecordingTest.doBasicRecordingByProfile(RecordingTes
t.java:1615)
E TestRunner:   at
android.hardware.camera2.cts.RecordingTest.basicRecordingTestByCamera(RecordingTe
st.java:1581)
E TestRunner:   at
android.hardware.camera2.cts.RecordingTest.doBasicRecording(RecordingTest.java:20
1)
E TestRunner:   at
android.hardware.camera2.cts.RecordingTest.doBasicRecording(RecordingTest.java:16
6)
E TestRunner:   at
android.hardware.camera2.cts.RecordingTest.testIntermediateSurfaceRecording(Recor
dingTest.java:257)
```

5.2.3.3 解决办法

上述帧率失败项，从log看基本都是帧率不够导致。

`sdk/hardware/rockchip/camera$` 目录看下是否有如下修改，没有的话手动打上，然后替换对应库；

```
commit a964c42b8dd34fa918ec215428287ee105ecb201
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Fri Nov 15 16:44:43 2024 +0800

    AAL: CameraStream: enable FPS print for debug
```

```
    Type: Function
    Redmine ID: # N/A
    Associated modifications: N/A
    Test: logcat | grep FPS

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: I9c89f17f57408bc7a8ed55029d826655b2cd75da
    commit a964c42b8dd34fa918ec215428287ee105ecb201
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Fri Nov 15 16:44:43 2024 +0800

    AAL: CameraStream: enable FPS print for debug

    Type: Function
    Redmine ID: # N/A
    Associated modifications: N/A
    Test: logcat | grep FPS

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: I9c89f17f57408bc7a8ed55029d826655b2cd75da

diff --git a/AAL/CameraStream.cpp b/AAL/CameraStream.cpp
index 3ba2666..411e3db 100644
--- a/AAL/CameraStream.cpp
+++ b/AAL/CameraStream.cpp
@@ -97,11 +97,11 @@ void CameraStream::showDebugFPS(int streamType)
                  (double)(1000000000)) / (double)diff;
         switch(streamType) {
             case STREAM_PREVIEW:
-                LOGI("%s: Preview FPS : %.4f: mFrameCount=%d", __func__, fps,
mFrameCount);
+                ALOGI("%s: Preview FPS : %.4f: mFrameCount=%d", __func__, fps,
mFrameCount);
                 LOGI("%s, mPendingRequests size=%zu", __func__,
mPendingRequests.size());
                 break;
             case STREAM_VIDEO:
-                LOGI("%s: Video FPS : %.4f", __func__, fps);
+                ALOGI("%s: Video FPS : %.4f", __func__, fps);
                 break;
             default:
                 break;
```

切换到录像模式，点击录像，log查看帧率是否能到30FPS

- 如果到不了，说明摄像头本身数据就有问题，另提redmine排查下驱动和效果文件配置情况。
- 如果能到，将测试机器放置到光照好的地方，不能遮挡摄像头测试。

5.2.4 SurfaceView相关

5.2.4.1 测试项

```
android.hardware.camera2.cts.SurfaceViewPreviewTest#testSurfaceSet[1]
```

5.2.4.2 报错日志

```
Wanted *at least* 30 times:
-> at
android.hardware.camera2.cts.SurfaceViewPreviewTest.verifyCaptureResults(SurfaceV
iewPreviewTest.java:891)
But was 28 times:
-> at
android.hardware.camera2.impl.CameraCaptureSessionImpl$1.lambda$onCaptureStarted$
0(CameraCaptureSessionImpl.java:650)

logcat中详细报错日志：
E Camera3-OutputStream: getBufferLockedCommon: Stream 13: Can't get next output
buffer from buffer manager: Function not implemented (-38)
E Camera3-BufferManager: getBufferForStream: bufferCount (18446744073709551615)
exceeds the max allowed buffer count (9) of this stream set
E Camera3-OutputStream: getBufferLockedCommon: Stream 13: Can't get next output
buffer from buffer manager: Function not implemented (-38)
E Camera3-BufferManager: getBufferForStream: bufferCount (18446744073709551615)
exceeds the max allowed buffer count (9) of this stream set
E Camera3-OutputStream: getBufferLockedCommon: Stream 13: Can't get next output
buffer from buffer manager: Function not implemented (-38)
```

5.2.4.3 解决办法

`sdkhardware/rockchip/camera_aidl$` 目录添加如下修改，如没有更新SDK或找FAE获取

```
commit d22ac4f020a7499e50ef8e120d77cdcdff837f6a
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Wed Aug 7 20:32:46 2024 +0800

    device: fix cts stream usage issue

    Test: CtsCameraTestCases

 android.hardware.camera2.cts.SurfaceViewPreviewTest#testSurfaceSet[1]

    Type: Fix
    Redmine ID: #500232
    Associated modifications: N/A
    Test: camera cts

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: Iba303389d5c0d25a6384a421a9b3a20cf198b045
```

5.2.5 Camera feature相关

5.2.5.1 测试项

```
android.app.cts.SystemFeaturesTest#testCameraFeatures
```

5.2.5.2 报错日志

```
java.lang.AssertionError. PackagelManager#hasSystemFeature should NOT return true
for android.hardware.camera
```

5.2.5.3 解决办法

从日志分析是板子没有后置摄像头，但配置了后置的feature，需要去掉；参考如下修改：

```
diff --git a/modules/camera_aidl.mk b/modules/camera_aidl.mk
index 5c18a5be..726b35b8 100644
--- a/modules/camera_aidl.mk
+++ b/modules/camera_aidl.mk
@@ -51,7 +51,6 @@ PRODUCT_COPY_FILES += \
   endif

   PRODUCT_COPY_FILES += \
  -
frameworks/native/data/etc/android.hardware.camera.xml:$(TARGET_COPY_OUT_VENDOR)/
etc/permissions/android.hardware.camera.xml \

frameworks/native/data/etc/android.hardware.camera.front.xml:$(TARGET_COPY_OUT_VE
NDOR)/etc/permissions/android.hardware.camera.front.xml
```

5.2.6 时间戳相关

5.2.6.1 测试项

```
android.hardware.camera2.cts.ReadoutTimestampTest#testReadoutTimestamp
```

5.2.6.2 报错日志

```
java.lang.Exception: There were 15 errors:
  java.lang.Throwable(Test failed for camera 0: Camera 0timestampBase 0:
readoutTime (249857275125000) - captureStart time (249857272121000) should be
&gt;= exposureTime (4193000))
  java.lang.Throwable(Test failed for camera 0: Camera 0timestampBase 0:
readoutTime (249857312509000) - captureStart time (249857308316000) should be
&gt;= exposureTime (5849000))
  java.lang.Throwable(Test failed for camera 0: Camera 0timestampBase 1:
readoutTime (249857423911000) - captureStart time (249857418062000) should be
&gt;= exposureTime (15998000))
```

5.2.6.3 解决办法

`sdkhardware/rockchip/camera_aidl$` 目录添加如下修改，如没有更新SDK或找FAE获取

```
commit 3ae91a564ad82c41bee71ebf2e65ea24fec853d0
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Mon Aug 12 20:23:40 2024 +0800

    device: fix mipi camera cts readoutTimestamp issue

    Test: CtsCameraTestCases

 android.hardware.camera2.cts.ReadoutTimestampTest#testReadoutTimestamp

    Type: Fix/Function
    Redmine ID: #500232
    Associated modifications: gerrit links
    Test: camera cts

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: I0f3fef0cb576f37fe004006b844adb349987c4df
```

5.2.7 测试包相关测试项

5.2.7.1 测试项

```
android.hardware.camera2.cts.CameraGPURecordingTest#testCameraGpuEncoderPath[1]
```

5.2.7.2 报错日志

```
junit.framework.AssertionFailedError: Camcorder profile should not be null
    at junit.framework.Assert.fail(Assert.java:50)
    at junit.framework.Assert.assertTrue(Assert.java:20)
    at junit.framework.Assert.assertNotNull(Assert.java:218)
    at
android.hardware.camera2.cts.CameraGPURecordingTest.prepareEncoder(CameraGPURecor
dingTest.java:174)
    at
android.hardware.camera2.cts.CameraGPURecordingTest.testCameraGpuEncoderPath(Came
raGPURecordingTest.java:106)
```

5.2.7.3 问题分析

```
经试验media_profiles_V1_0.xml打开720P录像支持后；
如果camera3_profiles.xml中配置后摄则能pass，配置前置则报错
junit.framework.AssertionFailedError: Camcorder profile should not be null
android.hardware.camera2.cts.CameraGPURecordingTest#testCameraGpuEncoderPath[1]
问题原因：
CTS测试代码：
14_Android_14.0_GMS\cts\tests\camera\src\android\hardware\camera2\cts\CameraGPURe
cordingTest.java
中通过如下获取测试逻辑：
        CamcorderProfile profile720p =
CamcorderProfile.get(CamcorderProfile.QUALITY_720P);
        assertNotNull("Camcorder profile should not be null", profile720p);
但实现的地方会判断是否为后摄，如果不是，就是NULL；
如果只有一个前置，就肯定是NULL，测试逻辑有问题；
14_Android_14.0_GMS\frameworks\base\media\java\android\media\CamcorderProfile.jav
a
    /**
     * Returns the default camcorder profile at the given quality level for the
first back-facing
     * camera on the device. If the device has no back-facing camera, this
returns null.
     * @param quality the target quality level for the camcorder profile
     * @see #get(int, int)
     * @deprecated Use {@link #getAll} instead
     */
    public static CamcorderProfile get(int quality) {
        int numberOfCameras = Camera.getNumberOfCameras();
        CameraInfo cameraInfo = new CameraInfo();
        for (int i = 0; i < numberOfCameras; i++) {
            Camera.getCameraInfo(i, cameraInfo);
            if (cameraInfo.facing == CameraInfo.CAMERA_FACING_BACK) {
                return get(i, quality);
            }
        }
        return null;
}
```

5.2.7.4 解决办法

经确认Android14.0-R4的CTS测试套件上述代码流程有问题，R5-Google更新了测试逻辑就正常了；

对应套件更新提交点

```
From 97c3d42e76661e66832e1d082c9cdafc09ff4a1d Mon Sep 17 00:00:00 2001
From: Bharatt Kukreja <bkukreja@google.com>
Date: Thu, 27 Jul 2023 21:17:10 +0000
Subject: [PATCH] Run test only for cameras which have the given camcorder profile

Currently if the test is run and the camera id does not have the given
camcorder profile (720p), prepareEncoder() fails the assertion to have a
non-null camcorder profile. Make sure to only run the test for camera
ids which have the profile.

Test: atest CameraGPURecordingTest.java

Bug: 292216589
Bug: 292216589

Merged-In: I58fe97c64b1e440a9900c93f0366ad9af5dd828d
Change-Id: I58fe97c64b1e440a9900c93f0366ad9af5dd828d
(cherry picked from commit 09c2b077b463b0eaaae4e7c3a2372ab65a73a1b1)
```

5.2.8 权限相关

5.2.8.1 测试项

```
android.permission3.cts.CameraMicIndicatorsPermissionTest#testMicIndicatorWithMan
ualFinishOpStillShows
```

5.2.8.2 报错日志

```
E TestRunner: failed:
testMicIndicatorWithManualFinishOpStillShows(android.permission3.cts.CameraMicInd
icatorsPermissionTest)
E TestRunner: ----- begin exception -----
E TestRunner: org.junit.ComparisonFailure: expected:<[Success]> but was:<[Failure
[DELETE_FAILED_INTERNAL_ERROR]]>
E TestRunner:   at org.junit.Assert.assertEquals(Assert.java:117)
E TestRunner:   at org.junit.Assert.assertEquals(Assert.java:146)
E TestRunner:   at
android.permission3.cts.CameraMicIndicatorsPermissionTest.uninstall(CameraMicIndi
catorsPermissionTest.kt:139)
E TestRunner:   at
android.permission3.cts.CameraMicIndicatorsPermissionTest.tearDown(CameraMicIndic
atorsPermissionTest.kt:197)
```

5.2.8.3 解决办法

根据报错信息看下测试逻辑，卸载应用"android.permissionui.cts.appthataccessescameraandmic"时失败，需使用新PAB包测试；

5.2.9 FOV相关

5.2.9.1 失败项

```
CTS-Verify Camera FOV Calibration
```

5.2.9.2 解决办法

参考如下步骤修改FOV相关参数值。

`sdk/frameworks/av/services/camera/libcameraservice/api1/client2/Parameters.cpp`

中调用如下函数计算FOV：

```
res = calculatePictureFovs(&horizFov, &vertFov);
 FOV计算公式：
/**
 \* Basic field of view formula is:
 \*  angle of view = 2 * arctangent ( d / 2f )
 \* where d is the physical sensor dimension of interest, and f is
 \* the focal length. This only applies to rectilinear sensors, for focusing
 \* at distances >> f, etc.
 */
```

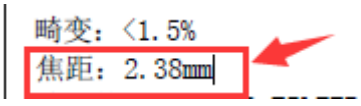
需要修改`camera3_profiles.xml`中的字段：

```
<lens.info.availableFocalLengths value="x.xx"/>
<sensor.info.physicalSize value="x.xx,x.xx"/> <!-- 1600x1.75um 1200x1.75um -->
```

上述两个值获取方法：

- lens.info.availableFocalLengths value

根据模组规格书提供,例如GC2145的一个模组规格书：



- sensor.info.physicalSize value

参考Sensor的Datasheet，例如GC2145，Pixel Size 1.75μm x 1.75μm

Pixel Size	1.75μm x 1.75μm
Active pixel array	1616 x 1232

```
则physicalSize value 中
hori = 1600x1.75um = 2.8 ;
vertical_size = 1200x1.75um =2.1;
<sensor.info.physicalSize value=" 2.8,2.1"/> <!-- 1600x1.75um 1200x1.75um -->
根据FOV计算公式： horizFov = arctan (2.8/2.38) = 49.6度;
注意：
如果实际CTS-Verify测量所需的值为：56度；则需要反推修改availableFocalLengths
因为physicalSize value是固定的，则根据FOV计算公式，
反推出availableFocalLengths值 = 2.8/tan((56°) = 2.8/1.48 = 1.89;
则可以上述公式，反推出修改成<lens.info.availableFocalLengths value="1.89/>即可。
```

5.3 Android 15.0 MIPI Camera CTS 常见失败项处理办法

5.3.1 STS相关测试项

5.3.1.1 测试项

```
StsHostTestCases android.security.sts.Poc20_01#testPocCVE_2019_14104
```

5.3.1.2 报错日志

```
The device has unexpectedly rebooted (41597 seconds after last recorded boot
time, bootreason: kernel_panic,oops:_fatal_exception)
expected to be less than: 1744204232
but was                 : 1744245819
    at
com.android.sts.common.tradefed.testtype.SecurityTestCase.tearDown(SecurityTestCa
se.java:160)

重启前kernel日志：
 pc : __pi_strlen+0x124/0x150
 lr : strstr+0x34/0x84
 sp : ffffffc014e1bb10
 x29: ffffffc014e1bb10 x28: 0000000000000000 x27: 0000000000000000
 x26: ffffff810c41b688 x25: 0000000000000001 x24: ffffff810c41b515
 x23: 0000001800000103 x22: 0000000000000001 x21: 0000000000000006
 x20: ffffffc001a13922 x19: 0000001800000103 x18: ffffffc010255060
 x17: 00000000fffe8f1f x16: 00000000fffe8f1f x15: 0000007fe705b048
 x14: 0000000000000000 x13: 0000000000000000 x12: 0000007fe705b048
 x11: 0000000080885658 x10: 0000001800000103 x9 : ffffff81daca4b00
 x8 : 0101010101010101 x7 : 7f7f7f7f7f7f7f7f x6 : 746d68736d6e61ff
 x5 : 0000000000000006 x4 : 0000000000000003 x3 : 756e69746e6f6300
 x2 : 307073696b727f7f x1 : 0000001800000100 x0 : 0000001800000103
 Call trace:
  __pi_strlen+0x124/0x150
  sditf_ioctl+0x570/0x810 [video_rkcif]
  subdev_do_ioctl_lock+0xcb0/0xf44
  video_usercopy+0x44c/0x98c
  subdev_ioctl+0x18/0x28
  v4l2_ioctl+0x6c/0x84
  __arm64_sys_ioctl+0xa8/0xe4
```

```
 invoke_syscall+0x58/0x11c
 el0_svc_common+0xb4/0xf4
 do_el0_svc+0x2c/0xb0
 el0_svc+0x2c/0x90
 el0t_64_sync_handler+0x68/0xb4
 el0t_64_sync+0x1a4/0x1a8
Code: fa4008a0 54fffde0 17ffffe6 927cec01 (a9400c22)
---[ end trace 0000000000000000 ]---
Kernel panic - not syncing: Oops: Fatal exception
```

5.3.1.3 解决办法

kernel添加如下提交，如没有找FAE获取

```
commit 136c9e13ceae281d48c89a3c1b87f439de3405f2
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Mon Jan 13 09:53:56 2025 +0800

    media: rockchip: isp: fix android sts CVE_2019_14104 failed issue

     fix test android.security.sts.Poc20_01#testPocCVE_2019_14104 cause
     kernel panic issue

     The issue was reported in https://redmine.rock-chips.com/issues/531294
     The kernel crash log:
     [ T7364] Unable to handle kernel paging request at virtual address
     0000001800000100
     ...
     pc : __pi_strlen+0x124/0x150
     lr : strstr+0x34/0x84
     ...

     Call trace:
     __pi_strlen+0x124/0x150
     sditf_ioctl+0x570/0x810 [video_rkcif]
     subdev_do_ioctl_lock+0xcb0/0xf44
     video_usercopy+0x44c/0x98c
     subdev_ioctl+0x18/0x28
     v4l2_ioctl+0x6c/0x84
     __arm64_sys_ioctl+0xa8/0xe4
     invoke_syscall+0x58/0x11c
     el0_svc_common+0xb4/0xf4
     do_el0_svc+0x2c/0xb0
     el0_svc+0x2c/0x90
     el0t_64_sync_handler+0x68/0xb4
     el0t_64_sync+0x1a4/0x1a8
     Code: fa4008a0 54fffde0 17ffffe6 927cec01 (a9400c22)
     ---[ end trace 0000000000000000 ]---

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: I26f2da26d9aec7b2f1dfa93c1165d7579b89f3ba
```

5.3.2 CtsVirtualDevicesCameraTestCases 相关

5.3.2.1 测试项

```
android.virtualdevice.cts.camera.VirtualCameraNdkTest#getCameraIds_virtualDevice
android.virtualdevice.cts.camera.VirtualCameraNdkTest#availabilityCallbacks_virtu
alDevice
```

5.3.2.2 报错日志

```
  <Module name="CtsVirtualDevicesCameraTestCases" abi="arm64-v8a" runtime="8063"
done="true" pass="0" total_tests="2">
    <TestCase name="android.virtualdevice.cts.camera.VirtualCameraNdkTest">
      <Test result="fail" name="getCameraIds_virtualDevice">
        <Failure message="missing (1): 1">
          <StackTrace>missing (1): 1
---
expected   : [1]
but was    : []
    at
android.virtualdevice.cts.camera.VirtualCameraNdkTest.getCameraIds_virtualDevice(
VirtualCameraNdkTest.java:148)</StackTrace>
        </Failure>
      </Test>
      <Test result="fail" name="availabilityCallbacks_virtualDevice">
        <Failure message="Wanted but not invoked:">
          <StackTrace>Wanted but not invoked:
mMockAvailabilityCallback.onCameraAvailable(
    "1"
);
-&amp;gt; at
android.virtualdevice.cts.camera.VirtualCameraNdkTest.availabilityCallbacks_virtu
alDevice(VirtualCameraNdkTest.java:164)
Actually, there were zero interactions with this mock.

    at
android.virtualdevice.cts.camera.VirtualCameraNdkTest.availabilityCallbacks_virtu
alDevice(VirtualCameraNdkTest.java:164)</StackTrace>
        </Failure>
      </Test>
    </TestCase>
  </Module>
```

5.3.2.3 问题分析

```
测试代码位置：
sdk\cts\tests\tests\virtualdevice\camera\src\android\virtualdevice\cts\camera\Vir
tualCameraNdkTest.java中getCameraIds_virtualDevice

测试失败问题原因：
List<String> ndkCameras = Arrays.asList(nativeCameraManager.getCameraIds());
```

获取到的List<String> ndkCameras size为0，导致
assertThat(ndkCameras).containsExactly(FRONT_CAMERA_ID);失败；

nativeCameraManager.getCameraIds()调用流程为：
---------------------->
Sdk\meworks\av\camera\ndk\impl\ACameraManager.cpp□CameraManagerGlobal::getCameraIdList(
---------□
CameraManagerGlobal::onStatusChangedLocked()
---------------》
CameraManagerGlobal::supportsCamera2ApiLocked()
---------□
sdk\frameworks\av\services\camera\libcameraservice\CameraService.cpp—
CameraService::supportsCameraApi
--------- CameraService::getDeviceVersion()----------》
sdk\frameworks\av\services\camera\libcameraservice\common\CameraProviderManager.cpp
------------------□ CameraProviderManager::getHighestSupportedVersion()---□
去匹配CameraID;异常，从而失败；
CameraId匹配异常原因：
是virtual_camera注册的cameraID为：v0/1/1_1000+deviceId值；
注册代码逻辑在：
sdk\frameworks\av\services\camera\virtualcamera\VirtualCameraService.cc中
createCameraId(const int32_t deviceId)
注册上去的CameraId名称为：
kCameraIdPrefix + std::to_string(deviceId) + "_" +
std::to_string(sNextIdNumericalPortion++);
其中：
kCameraIdPrefix[] = "v";
static std::atomic_int sNextIdNumericalPortion{1000};
deviceId是根据虚拟设备的ID；

而下发匹配的CameraID为：
Sdk\frameworks\av\camera\ndk\impl\ACameraManager.cpp
中mCameraService->addListener(mCameraServiceListener, &cameraStatuses);里面
hardware::CameraStatus的cameraId，打印获取到的为0/1

综合分析：
CtsVirtualDevicesCameraTestCases测试代码为Google 原生代码；
虚拟摄像头的实现逻辑：在 sdk/frameworks/av/services/camera/virtualcamera$ framework层，vendor改不了；

总结：CtsVirtualDevicesCameraTestCases为Android15.0 新增测试项,测试失败为测试包本身测试逻辑问题；
平板能测试过，是因为刚好平板有前后摄像头注册的cameraID String为：0/1刚好匹配上了.

5.3.2.4 解决办法

目前这两项失败项属于豁免。

5.3.3 闪光灯相关测试项

5.3.3.1 测试项

```
android.hardware.camera2.cts.CaptureRequestTest#testFlashControl[1]
```

5.3.3.2 报错日志

```
E TestRunner: failed: testFlashControl[1]
(android.hardware.camera2.cts.CaptureRequestTest)
E TestRunner: ----- begin exception -----
E TestRunner: java.lang.Exception: There were 22 errors:
E TestRunner:   java.lang.Throwable(Test failed for camera 0: Flash mode result
must be SINGLE (expected = 1, actual = 0) )
E TestRunner:   java.lang.Throwable(Test failed for camera 0: Flash mode result 0
must be TORCH (expected = 2, actual = 0) )
E TestRunner:   java.lang.Throwable(Test failed for camera 0: Flash mode result 1
must be TORCH (expected = 2, actual = 0) )
E TestRunner:   java.lang.Throwable(Test failed for camera 0: Flash mode result 2
must be TORCH (expected = 2, actual = 0) )
E TestRunner:   java.lang.Throwable(Test failed for camera 0: Flash mode result 3
must be TORCH (expected = 2, actual = 0) )
```

5.3.3.3 解决办法

`sdkhardware/rockchip/camera$` 目录添加如下修改，如没有更新SDK或找FAE获取

```
commit c65c8cca26fc0c9a5229ba6e1943be6184ea35ba
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Wed Jan 15 10:13:48 2025 +0800

    psl: rkisp2: fix cts Flash mode result not correct issue

    Type: Fix
    Redmine ID: # N/A
    Associated modifications:
    https://10.10.10.29/c/android/rk/hardware/rk29/camera_v3/+/233300
    Test: CtsCameraTestCases
    android.hardware.camera2.cts.CaptureRequestTest#testFlashControl[1]

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: Ie5cd70b04fd7ce7e0f41380e7f7b63b707a6652c
```

5.3.4 其他已修复测试项

5.3.4.1 测试项

```
android.hardware.camera2.cts.MultiViewTest#testTextureImageWriterReaderOperation[
1]
```

5.3.4.2 报错日志

```
E TestRunner: failed: testTextureImageWriterReaderOperation[1]
(android.hardware.camera2.cts.MultiViewTest)
E TestRunner: ----- begin exception -----
E TestRunner: junit.framework.AssertionFailedError: Bitmap difference exceeds
threshold: diff 84.870528 > threshold 0.100000
E TestRunner:   at junit.framework.Assert.fail(Assert.java:50)
E TestRunner:   at junit.framework.Assert.assertTrue(Assert.java:20)
E TestRunner:   at
android.hardware.camera2.cts.MultiViewTest.testTextureImageWriterReaderOperation(
MultiViewTest.java:975
```

5.3.4.3 解决办法

`sdkhardware/rockchip/camera$` 目录添加如下修改，如没有更新SDK或找FAE获取

```
commit e9cee7931e18e81efb3ddcf19bae5a9bd99d07ef
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Thu Jan 23 17:40:21 2025 +0800

    rkisp2: runtime do rga handle map for Multi memcopy case to fix cts issue

    in following case, use init mapped rga handle cause preview green issue,
    need runtime import buffer_fd to get correct rga handle

    Type: Fix
    Redmine ID: # N/A
    Associated modifications:
    https://10.10.10.29/c/android/rk/hardware/rk29/camera_v3/+/225116
    Test: CtsCameraTestCases

android.hardware.camera2.cts.MultiViewTest#testTextureImageWriterReaderOperation[
1]

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: I75261e187247d25f50f79e34c775020ef5bdebd1
```

5.3.5 CTS-Verifier BYOD Managed Provisioning相关测试项

5.3.5.1 测试项 1

```
Camera support cross profile video capture (with extra output path)
```

5.3.5.2 报错日志

```
E AndroidRuntime: FATAL EXCEPTION: main
E AndroidRuntime: Process: com.android.camera2, PID: 6916
E AndroidRuntime: java.lang.RuntimeException: No valid video file descriptor
E AndroidRuntime: at
com.android.camera.VideoModule.initializeRecorder(VideoModule.java:1365)
E AndroidRuntime: at
com.android.camera.VideoModule.-$$Nest$minitializeRecorder(VideoModule.java:0
E AndroidRuntime: at
com.android.camera.VideoModule$13.onStorageUpdateDone(VideoModule.java:1582)
E AndroidRuntime: at
com.android.camera.CameraActivity$23.onPostExecute(CameraActivity.java:2706)
E AndroidRuntime: at
com.android.camera.CameraActivity$23.onPostExecute(CameraActivity.java:2690)
E AndroidRuntime: at android.os.AsyncTask.finish(AsyncTask.java:771)
E AndroidRuntime: at android.os.AsyncTask.-$$Nest$mfinish(Unknown Source:0)
E AndroidRuntime: at
android.os.AsyncTask$InternalHandler.handleMessage(AsyncTask.java:788)
```

5.3.5.3 解决办法

`sdk/packages/apps/Camera2$` 目录添加如下修改，如没有更新SDK或找FAE获取

```
commit cbf416df82b90e8cb26cf08d91faf36a37b776db
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Thu Mar 13 17:36:47 2025 +0800

    Camera2: fix ctsv camera intent fail

    failed log:
      java.lang.RuntimeException: No valid video file descriptor
        at
com.android.camera.VideoModule.initializeRecorder(VideoModule.java:1365)
        at
com.android.camera.VideoModule.-$$Nest$minitializeRecorder(VideoModule.java:0)
        at
com.android.camera.VideoModule$13.onStorageUpdateDone(VideoModule.java:1582)
        at
com.android.camera.CameraActivity$23.onPostExecute(CameraActivity.java:2706)
        at
com.android.camera.CameraActivity$23.onPostExecute(CameraActivity.java:2690)

    Type: Fix
    Redmine ID: # 540432
    Associated modifications: N/A
    Test: run ctsv camera intent test
```

```
    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: Ibdbc3df569dc78ada6d1b4fc15303d30522b369
```

5.3.5.4 测试项 2

```
Camera support cross profile video capture (without extra output path)
```

5.3.5.5 报错日志

```
 失败描述：
 Camera support cross profile video capture (without extra output path) fail,实际抓不到画面，无法播放

 报错日志：
 V CAM_CameraAppUI: SurfaceTexture is destroyed
 V CAM_VideoModule: Skip stopPreview since it's not mPreviewing
 D CAM_VideoUI: surfaceTexture is destroyed
 D CoreBackPreview: Window{a48008d u0 com.android.camera2/com.android.camera.VideoCamera}: Setting back callback null
```

5.3.5.6 解决办法

`sdk/packages/apps/Camera2$` 目录添加如下修改，如没有更新SDK或找FAE获取

```
commit 37d5a641010dda1981b3d74504b950dcea626c93
Author: president.chao <president.chao@rock-chips.com>
Date:   Tue Mar 25 14:24:48 2025 +0800

    Camera2 :Fix Camera support cross profile video capture (without extra output path) fail

    Type: Fix
    Redmine ID: #540457
    Associated modifications: N/A
    Test: N/A

    Change-Id: I57faaf227a7d7e275ec4a07aef3bc219b5b19af2
    Signed-off-by: president.chao <president.chao@rock-chips.com>
```

5.4 不支持Camera设备，常见失败项处理办法

5.4.1 CTS-Verifier 相关BYOD Managed Provisioning

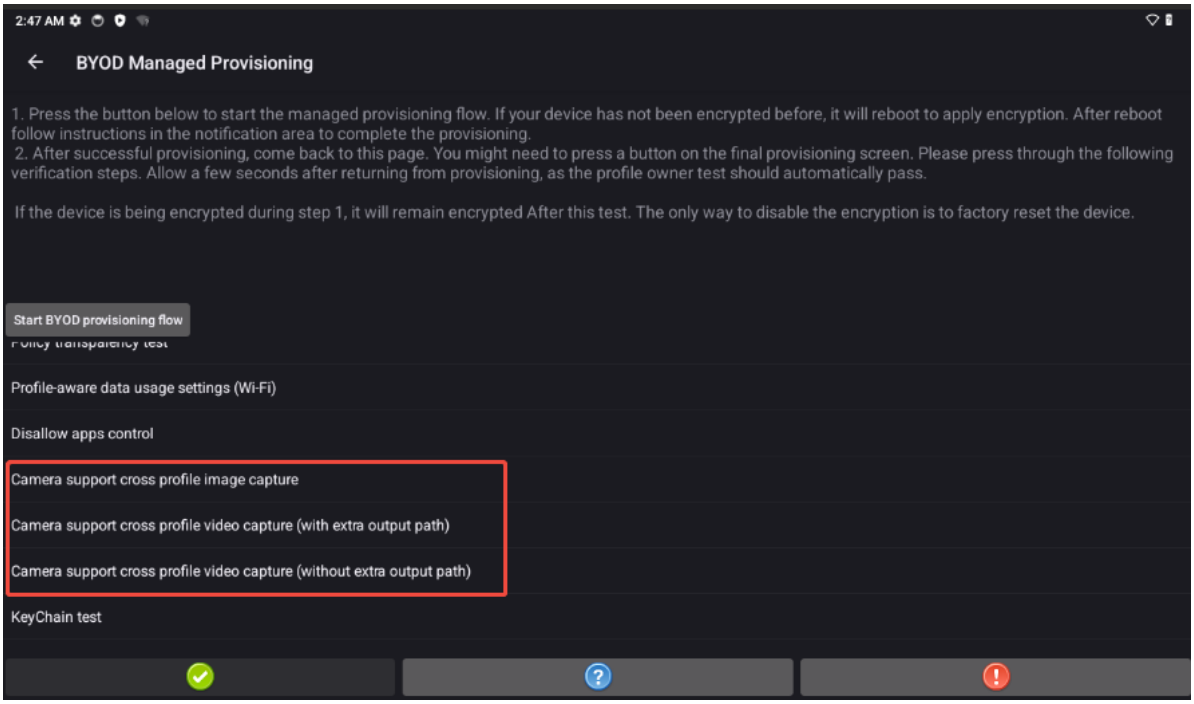
若机器没有Camera则CTS-Verifier要求不能出现下面3个测试项 。

5.4.1.1 测试项

```
1. Camera support cross profile image capture
2. Camera support cross profile video capture (with extra output path)
3. Camera support cross profile video capture (without extra output path)
```

5.4.1.2 报错日志

不支持camera的设备，正常是没有这三条camera的case的。



5.4.1.3 问题分析

1. 使用如下命令确认camera相关feature是否都已经去掉：

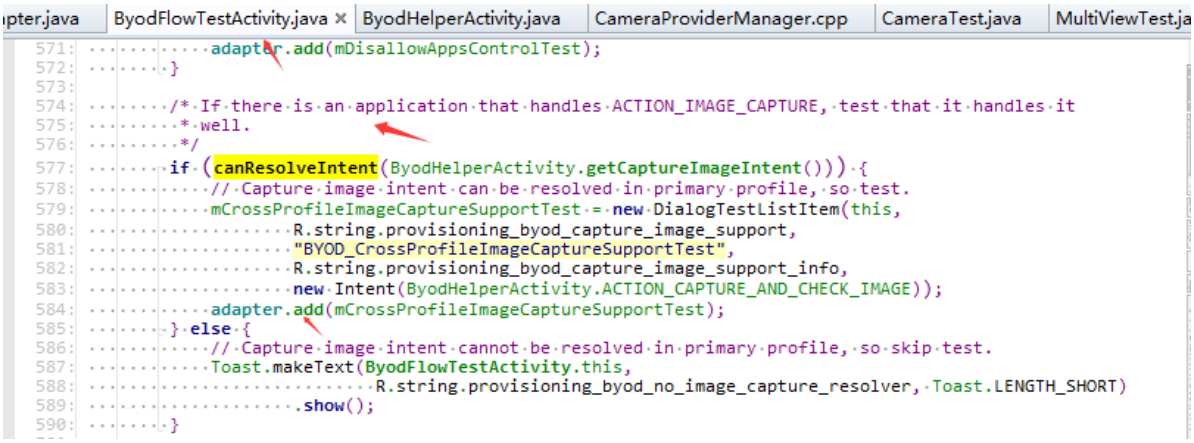
```
arm64:/ # pm list features | grep cam
```

2. 查看分析测试代码逻辑

测试代码位置为：

```
sdk\cts\apps\CtsVerifier\src\com\android\cts\verifier\managedprovisioning\ByodFlowTestActivity.java
```

测试逻辑是有应用支持ACTION_IMAGE_CAPTURE就会添加该测试项；



所以是需要将Camera相关应用删除才行；

5.4.1.4 解决办法

移除Camera相关feature申明，移除Camera2应用:

`sdk/device/rockchip/common$`目录添加如下修改，如没有找FAE获取

```
commit cd1fba57979e5e6c9162b97762fc41cc8c99b827 (HEAD)
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Thu Aug 7 15:31:29 2025 +0800

    fix ctsv no camera failed issue

    No camera no need to compile Camera2 APk,
    it causes CTS-Verifier BYOD Managed Provisioning
    following test cases still exists:

    1. Camera support cross profile image capture
    2. Camera support cross profile video capture (with extra output path)
    3. Camera support cross profile video capture (without extra output path)

    Type: Fix
    Redmine ID: # 570773
    Associated modifications: N/A
    Test: CTS-Verifier -- BYOD Managed Provisioning

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: I36f61d010809afebaceab955439d1f4756313aad
```

或者参考如下修改：

```
Android_SDK/build/make$ git diff target/product/handheld_product.mk
PRODUCT_PACKAGES += \
Browser2 \
Calendar \
- Camera2 \
Android_SDK/device/rockchip/common$ git diff
./overlay/frameworks/base/core/res/res/values/config.xml
<!-- Whether this device is supporting the software microphone toggle -->
<bool name="config_supportsMicToggle">true</bool>
<!-- Whether this device is supporting the camera toggle -->
- <bool name="config_supportsCamToggle">true</bool>
+ <bool name="config_supportsCamToggle">false</bool>
```

5.4.2 GtsInteractiveCredentialsTestCases模块

5.4.2.1 测试项

```
GtsCredentialManagerQrCodeTest#testCameraQRCode
```

5.4.2.2 报错日志

```
E TestRunner: failed:
testCameraQRCode(com.google.android.gts.credentials.interactive.GtsCredentialMana
gerQrCodeTest)
E TestRunner: ----- begin exception -----
E TestRunner: com.android.interactive.StepFailedException: Step
com.google.android.interactive.steps.credentials.LaunchDefaultCameraAppStep
marked as failed by tester
E TestRunner:   at com.android.interactive.Step.execute(Step.java:208)
E TestRunner:   at
com.google.android.gts.credentials.interactive.GtsCredentialManagerQrCodeTest.tes
tCameraQRCode(GtsCredentialManagerQrCodeTest.kt:66)
E TestRunner:   at java.lang.reflect.Method.invoke(Native Method)
E TestRunner:   at
org.junit.runners.model.FrameworkMethod$1.runReflectiveCall(FrameworkMethod.java:
59)
E TestRunner:   at
org.junit.internal.runners.model.ReflectiveCallable.run(ReflectiveCallable.java:1
2)
E TestRunner:   at
org.junit.runners.model.FrameworkMethod.invokeExplosively(FrameworkMethod.java:61
)
E TestRunner:   at
org.junit.internal.runners.statements.InvokeMethod.evaluate(InvokeMethod.java:17)
E TestRunner:   at
org.junit.internal.runners.statements.FailOnTimeout$CallableStatement.call(FailOn
Timeout.java:148)
E TestRunner:   at
org.junit.internal.runners.statements.FailOnTimeout$CallableStatement.call(FailOn
Timeout.java:142)
E TestRunner:   at java.util.concurrent.FutureTask.run(FutureTask.java:264)
E TestRunner:   at java.lang.Thread.run(Thread.java:1012)
```

```
E TestRunner: ----- end exception -----
I TestRunner: finished:
testCameraQRCode(com.google.android.gts.credentials.interactive.GtsCredentialMana
gerQrCodeTest)
I TestRunner: run finished: 1 tests, 1 failed, 0 ignored
```

5.4.2.3 问题分析

客户反馈无相机的设备，正常的话就不会弹出CameraQRCode这项的，这项是手动测试的，实验室他们看到弹出CameraQRCode测试界面就会直接判断为Fail的，所以测试失败。

客户反馈其他客户跳过这项测试，测试跳过的LOG如下：

```
I TestRunner: run started: 1 tests
I TestRunner: started:
testCameraQRCode(com.google.android.gts.credentials.interactive.GtsCredentialMana
gerQrCodeTest)
E TestRunner: assumption failed:
testCameraQRCode(com.google.android.gts.credentials.interactive.GtsCredentialMana
gerQrCodeTest)
E TestRunner: ----- begin exception -----
E TestRunner: org.junit.AssumptionViolatedException: got: <false>, expected: is
<true>
E TestRunner:     at org.junit.Assume.assumeThat(Assume.java:106)
E TestRunner:     at org.junit.Assume.assumeTrue(Assume.java:50)
E TestRunner:     at
com.google.android.gts.credentials.interactive.GtsCredentialManagerQrCodeTest.set
up(GtsCredentialManagerQrCodeTest.kt:44)
```

分析GtsInteractiveCredentialsTestCases.apk测试代码，查看上述跳过逻辑：

默认产品TV/Car/Watch 这几项会默认skip掉，以及SDK Version 设置 <=34 ，就assumption failed；

```
        this.context = context;
33      PackageManager packageManager = this.context.getPackageManager();
        Intrinsics.checkNotNullExpressionValue(packageManager, "getPackageManager(...)");
        this.packageManager = packageManager;
35      this.isTv = this.packageManager.hasSystemFeature("android.software.leanback");
36      this.isCar = this.packageManager.hasSystemFeature("android.hardware.type.automotive");
37      this.isWatch = this.packageManager.hasSystemFeature("android.hardware.type.watch");
    }

    @Before
40  public final void setup() {
41      Assume.assumeFalse("Device is automotive", this.isCar);
42      Assume.assumeFalse("Device is watch", this.isWatch);
43      Assume.assumeFalse("Device is TV", this.isTv);
44      Assume.assumeTrue(SdkLevel.isAtLeastV());
47      if (!GmsUtil.hasGmsCore()) {
48          Assume.assumeTrue("GMS Core is not installed", false);
        } else {
53          Assume.assumeTrue("GMS Core is CN build", !GmsUtil.isCnBuild());
        }
    }

    @GmsTest(requirement = "GMS-8.33-001/GMS-8.33-002/GMS-8.33-004")
    @RequireSdkVersion(min = 35)
    @Interactive
    @Test
    @RequireHandheldDevice
    @NotFullyAutomated(reason = "Tester needs to manually scan QR code with camera")
    @RequireNotCnGmsBuild
64  public final void testCameraQRCode() {
65      Step.execute(GrantPermissionForCameraAppStep.class);
66      Step.execute(LaunchDefaultCameraAppStep.class);
67      Step.execute(PointPhoneCameraToPasskeyQRCodeStep.class);
68      Step.execute(TriggerQRCodeURIActionStep.class);
    }
}
```

5.4.2.4 解决办法

从测试逻辑分析，如果SDK verison >= 35，这项肯定就会跑，所以测试人员手动点failed，肯定不行；

需实验室跟Google申请豁免就可以了。

6. ITS

6.1 ITS各测试项官方说明

ITS各测试项官方说明

6.2 ITS 测试报告查看

- 示例报告

单个测试场景示例报告汇总文件：
CameraITS_4_luokzb\cam_id_0\scene2_a中文件scene_test_summary.txt

```
Cam0 scene2_a
SKIP  test_auto_flash.py
SKIP  test_display_p3.py
SKIP  test_effects.py
PASS  test_format_combos.py
PASS  test_jpeg_quality.py
SKIP  test_num_faces.py
FAIL  test_preview_min_frame_rate.py
SKIP  test_reprocess_uv_swap.py
```

- 报告解读

```
Cam0 scene2_a
//Camera ID 0，测试scene2_a 场景
SKIP/PASS/FAIL
```

6.3 ITS常见failed项处理办法

6.3.1 帧率问题

6.3.1.1 测试项

```
scene2_a test_preview_min_frame_rate.py
```

6.3.1.2 报错日志

```
Traceback (most recent call last):
  File "/home/fih/USER/ITS_14/lib/python3.9/site-packages/mobly/base_test.py",
line 756, in exec_one_test
    test_method()
  File "/home/fih/USER/ITS_14/android-cts-verifier-14_r4-linux_x86-arm/android-
cts-verifier/CameraITS/tests/scene2_a/test_preview_min_frame_rate.py", line 133,
in test_video_min_frame_rate
    raise AssertionError('\n'.join(errors))
AssertionError: Preview frame rate was 30.001912873639593, expected to be 15,
ATOL: 2.
```

6.3.1.3 解决办法

调整json效果文件中帧率动态范围，支持从15-30动态波动。

RK3562平台示例：

```
gc02m1_KYT-8789-V10_default.json
"TimeDot":  [0, 0.03, 0.05, 0.05, 0.05, 0.05],
改成：
"TimeDot":  [0, 0.01, 0.01, 0.03, 0.067, 0.070],
```

RK3576平台示例：

```
ov16880_KYT-11379-V1_default.json中
"sw_aeT_time_dot":  [0, 0.04, 0.04, 0.04, 0.04, 0.04],
改成：
"sw_aeT_time_dot":  [0, 0.03, 0.03, 0.04, 0.067, 0.070],
```

6.3.2 闪光灯相关测试项

6.3.2.1 测试项

```
 scene_flash test_auto_flash.py
```

6.3.2.2 报错日志

```
DEBUG Flash frames Y mean: 149.1733
ERROR Exception occurred in test_auto_flash.
Traceback (most recent call last):
  File "/home/gmsits/miniconda3/envs/py310/lib/python3.10/site-
packages/mobly/base_test.py", line 783, in exec_one_test
    test_method()
  File "/home/gmsits/GMS_work/ITS/15/NEW-R2/android-cts-verifier-15_r2-linux_x86-
arm/android-cts-verifier/CameraITS/tests/scene_flash/test_auto_flash.py", line
143, in test_auto_flash
    raise AssertionError(f'mean FLASH-OFF: {mean_delta:.3f}, '
AssertionError: mean FLASH-OFF: 4.715, ATOL: 15
```

6.3.2.3 原因分析

```
ITS scene_flash test_auto_flash.py这个测试和正常切换分辨率的闪光点拍照流程不一致；
拍照时候没有默认下发：ANDROID_CONTROL_AE_PRECAPTURE_TRIGGER_START，
而是进入LOGD_CAP("forcely trigger ae precapture")流程，从而没有触发闪光灯导致失败。
```

6.3.2.4 解决办法

`sdk/hardware/rockchip/camera$` 目录添加如下修改，如没有更新SDK或找FAE获取

```
commit c220ca025b0350e82313782f52cbf476bbda8380
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Mon Dec 23 19:40:08 2024 +0800

    rkisp2: fix camera ctsv its test_auto_flash fail

    switch resolution picture need differ flash process,
    fix camera ctsv its scene_flash test_auto_flash.py

    Test: CTSV ITS
    scene_flash test_auto_flash.py

    Type: Fix
    Redmine ID: # 525721
    Associated modifications: N/A
    Test: ITS test

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: I674a5cc94bd0942dbfbe1dfb6a44fb3fac0cc643
```

6.3.3 方向flip/mirror相关测试项

6.3.3.1 测试项

```
scene3 test_flip_mirror.py
```

6.3.3.2 报错日志1

```
09-12 22:22:08.864 WARNING Warning: unable to find chart in scene!
Check camera distance and self-reported pixel pitch, focal length and hyperfocal
distance.
```

6.3.3.3 原因分析

原因分析：离太远了，导致测试卡没识别；

6.3.3.4 解决办法1

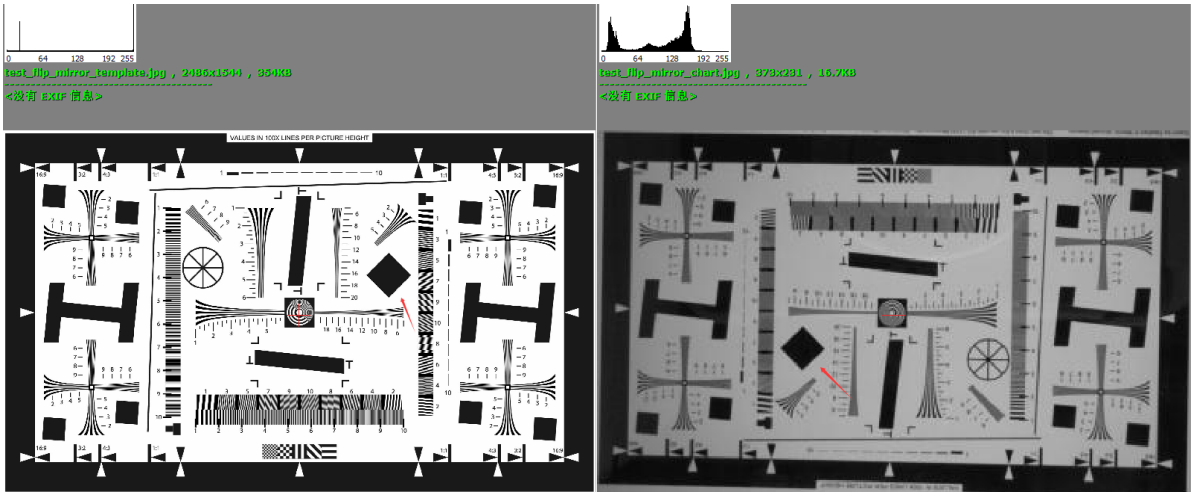
测试图位置放好，自测pass。

6.3.3.5 报错日志2

```
11-04 15:50:34.860 ERROR Exception occurred in test_flip_mirror.
Traceback (most recent call last):
  File "/home/gmsits/miniconda310/envs/A16/lib/python3.10/site-
packages/mobly/base_test.py", line 783, in exec_one_test
    test_method()
  File "/home/gmsits/GMS_work/ITS/16/R2/android-cts-verifier-16_r2-linux_x86-
arm/android-cts-verifier/CameraITS/tests/scene3/test_flip_mirror.py", line 175,
in test_flip_mirror
    test_flip_mirror_impl(cam, props, fmt, chart, first_api_level,
  File "/home/gmsits/GMS_work/ITS/16/R2/android-cts-verifier-16_r2-linux_x86-
arm/android-cts-verifier/CameraITS/tests/scene3/test_flip_mirror.py", line 128,
in test_flip_mirror_impl
    raise AssertionError(
AssertionError: Optimum orientation is rotate
```

6.3.3.6 原因分析

查看报告中保存的图片对比发现和template图像有flip/mirror异常；



6.3.3.7 解决办法2

1、首先确认camera3_profiles.xml中摄像头方向配置是否满足如下规则

```
竖屏机器：前置摄像头配置270°，后置摄像头配置90°；
横屏机器：前置摄像头配置180°，后置摄像头配置0°；
关键字：`<sensor.orientation value="0"/>`
如果不符合，先修改，sync后，推回板子确认方向；
```

修改后方向一般会差异180°

2、然后修改CMOS sensor驱动，调整方向正确

参考CMOS sensor的Datasheet，找到对应Flip、Mirror寄存器，使用i2c命令进行调整
例如：GC05A2 FLIP/MIRROR寄存器如下为：0x0101

Table 4-1 Mirror and Flip Information

Function	Register Address	Register Value	First Pixel
Normal	0x0101[1:0]	00	Gr
Horizontal mirror	0x0101[1:0]	01	R
Vertical Flip	0x0101[1:0]	10	B
Horizontal Mirror and Vertical Flip	0x0101[1:0]	11	Gb

```
i2ctransfer -y -f 4 w2@0x37 0x01 0x01 r1   //读取原有寄存器值
i2ctransfer -y -f 4 w3@0x37 0x01 0x01 0x00 //normal
i2ctransfer -y -f 4 w3@0x37 0x01 0x01 0x01 //mirror
i2ctransfer -y -f 4 w3@0x37 0x01 0x01 0x02 //flip
i2ctransfer -y -f 4 w3@0x37 0x01 0x01 0x03 //flip & mirror
```

调整符和方向的寄存器配置后，可以根据如下 两种情况修改驱动完成：

- CMOS sensor仅输出YUV格式数据，则直接修改驱动中初始化寄存器配置调整方向；
- CMOS sensor仅输出RAW格式数据，则还可以驱动实现V4L2标准ioctl调用V4L2_CID_HFLIP和V4L2_CID_VFLIP，从而使用IQ文件实现方向调整

3、示例解决方法

以输出RAW格式的前摄GC05A2驱动调整方向为例说明

- gc05a2添加V4L2_CID_HFLIP和V4L2_CID_VFLIP支持修改:

```
diff --git a/drivers/media/i2c/gc05a2.c b/drivers/media/i2c/gc05a2.c
index 233063592e03..bb4f7ab6ccc5 100644
--- a/drivers/media/i2c/gc05a2.c
+++ b/drivers/media/i2c/gc05a2.c
@@ -81,6 +82,10 @@
 #define GC05A2_REG_VTS_H       0x0340
 #define GC05A2_REG_VTS_L       0x0341

+#define GC05A2_FLIP_MIRROR_REG     0x0101
+#define MIRROR_BIT_MASK            BIT(0)
+#define FLIP_BIT_MASK          BIT(1)
```

```
+
 #define REG_NULL                0xFFFF

 #define OF_CAMERA_PINCTRL_STATE_DEFAULT    "rockchip,camera_default"
@@ -139,6 +144,9 @@ struct gc05a2 {
    struct v4l2_ctrl    *hblank;
    struct v4l2_ctrl    *vblank;
    struct v4l2_ctrl    *link_freq;
+   struct v4l2_ctrl    *h_flip;
+   struct v4l2_ctrl    *v_flip;
+   u8          flip_mirror;
    struct mutex        mutex;
    bool            streaming;
    unsigned int        lane_num;
@@ -1439,6 +1405,27 @@ static int gc05a2_set_ctrl(struct v4l2_ctrl *ctrl)
                    (ctrl->val + gc05a2->cur_mode->height)
                    & 0xff);
        break;
+   case V4L2_CID_HFLIP:
+       dev_dbg(&client->dev, "set mirror value 0x%x\n", ctrl->val);
+
+       if (ctrl->val)
+           gc05a2->flip_mirror |= MIRROR_BIT_MASK;
+       else
+           gc05a2->flip_mirror &= ~MIRROR_BIT_MASK;
+
+       ret |= gc05a2_write_reg(gc05a2->client, GC05A2_FLIP_MIRROR_REG,
+                   gc05a2->flip_mirror);
+       break;
+   case V4L2_CID_VFLIP:
+       dev_dbg(&client->dev, "set flip value 0x%x\n", ctrl->val);
+       if (ctrl->val)
+           gc05a2->flip_mirror |= FLIP_BIT_MASK;
+       else
+           gc05a2->flip_mirror &= ~FLIP_BIT_MASK;
+
+       ret |= gc05a2_write_reg(gc05a2->client, GC05A2_FLIP_MIRROR_REG,
+                   gc05a2->flip_mirror);
+       break;
 @@ -1501,6 +1488,11 @@ static int gc05a2_initialize_controls(struct gc05a2
*gc05a2)
                V4L2_CID_ANALOGUE_GAIN, GC05A2_AGAIN_MIN,
                GC05A2_AGAIN_MAX, GC05A2_AGAIN_STEP,
                GC05A2_AGAIN_DEFAULT);
+   gc05a2->h_flip = v4l2_ctrl_new_std(handler, &gc05a2_ctrl_ops,
+                     V4L2_CID_HFLIP, 0, 1, 1, 0);
+
+   gc05a2->v_flip = v4l2_ctrl_new_std(handler, &gc05a2_ctrl_ops,
+                     V4L2_CID_VFLIP, 0, 1, 1, 0);

    if (handler->error) {
        ret = handler->error;
```

- camera3_profiles.xml修改正确orientation

```
diff --git a/etc/camera/camera3_profiles_rk3576.xml
b/etc/camera/camera3_profiles_rk3576.xml
index 746dea9..2220141 100644
--- a/etc/camera/camera3_profiles_rk3576.xml
+++ b/etc/camera/camera3_profiles_rk3576.xml
@@ -1484,7 +1484,7 @@
             <sensor.baseGainFactor value="0,1"/> <!-- HAL may override this
value from CMC for RAW sensors -->
             <sensor.blackLevelPattern value="0,0,0,0"/>
             <sensor.maxAnalogSensitivity value="2400"/> <!-- HAL may override
this value from CMC for RAW sensors -->
-            <sensor.orientation value="90"/>
+            <sensor.orientation value="270"/>
             <sensor.profileHueSatMapDimensions value="0,0,0"/>
             <sensor.availableTestPatternModes value="OFF,COLOR_BARS"/>
```

- IQ文件调整Sensor_info中FLIP配置

```
diff --git a/iqfiles/isp39/gc05a2_KYT-11210-V2_default.json
b/iqfiles/isp39/gc05a2_KYT-11210-V2_default.json
index 4ad124c3..5d7d9403 100644
--- a/iqfiles/isp39/gc05a2_KYT-11210-V2_default.json
+++ b/iqfiles/isp39/gc05a2_KYT-11210-V2_default.json
@@ -111,7 +111,7 @@
                         }
                 },
                 "CISMinFps":    10,
-                "CISFlip":      0
+                "CISFlip":      3
         },
         "module_calib": {
                 "sensor_module":        {
```

6.3.4 摄像头3A相关测试项

6.3.4.1 测试项

```
scene8 test_ae_awb_regions.py
```

6.3.4.2 报错日志

```
09-12 11:25:22.006 ERROR Exception occurred in test_ae_awb_regions.
Traceback (most recent call last):
  File "/home/gmsits/miniconda310/envs/A16/lib/python3.10/site-
packages/mobly/base_test.py", line 783, in exec_one_test
    test_method()
  File "/home/gmsits/GMS_work/ITS/16/R2/android-cts-verifier-16_r2-linux_x86-
arm/android-cts-verifier/CameraITS/tests/scene8/test_ae_awb_regions.py", line
269, in test_ae_awb_regions
    _do_ae_check(light, dark, file_name_with_path)
  File "/home/gmsits/GMS_work/ITS/16/R2/android-cts-verifier-16_r2-linux_x86-
arm/android-cts-verifier/CameraITS/tests/scene8/test_ae_awb_regions.py", line 74,
in _do_ae_check
    raise AssertionError(
AssertionError: Luma change 0.030840987997693057 is less than the threshold: 1
```

6.3.4.3 原因分析

该测试项，需要API LEVEL 比Android15.0 的高，才会测试

`first_api_level >= its_session_utils.ANDROID15_API_LEVEL`

测试逻辑：
测试在不同的自动曝光 (AE) 和自动白平衡 (AWB) 区域进行预览录制时，RGB 和亮度值是否存在差异。该测试会录制 8 秒的预览录制内容，并对每个象限进行 AE 和 AWB 测光，每个象限各 2 秒。然后，该测试会从每个区域的预览录制中提取一个帧，并使用提取的帧执行以下 AE 和 AWB 检查：

AE 检查：验证对曝光度较低的区域进行帧测光时，与对曝光度较高的区域进行帧测光相比，亮度值是否增加了 1% 以上。该检查可验证：在对较暗区域进行测光时，图片会变亮。

AWB 检查：验证具有蓝色测光区域的帧中（图片平均 RGB 值的）红色与蓝色的比例是否比具有黄色测光区域的帧高出 2% 以上。该检查可验证：在对黄色（暖色）或蓝色（冷色）区域测光时，图像的 RGB 值是均衡的。



6.3.4.4 解决办法

`sdk/external/camera_engine_rkaiq$ $` 目录添加如下修改，如没有更新SDK或找FAE获取

```
commit 44e5a2e550e2783c65d7ea460b67380c9eec5696
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Wed Sep 24 14:13:32 2025 +0800

    hal_interface2.0: interface: fix camera ctsv its ae_regions test fail

    Test: CTSV ITS
    scene8 test_ae_awb_regions.py

    Type: Fix
    Redmine ID: # 579695
    Associated modifications: N/A
    Test: ITS test

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: I9d0ae30a4a2eb04a0859e214b7020c2e6772f823
```

6.3.5 其他已修复测试项

6.3.5.1 测试项

```
scene4 test_video_aspect_ratio_and_crop.py
```

6.3.5.2 报错日志

```
01-24 16:03:22.211 ERROR FoV test summary for quality: QVGA
01-24 16:03:22.211 ERROR Images failed in the FoV test:
01-24 16:03:22.211 ERROR Quality: QCIF FoV : 13.70, Ref FoV %: 7.75, TOL=15.0,
img: 176x144, ref: 2592x1944
01-24 16:03:22.211 ERROR Quality: 480P FoV : 12.66, Ref FoV %: 7.10, TOL=15.0,
img: 640x480, ref: 2592x1944
01-24 16:03:22.211 ERROR Quality: QVGA FoV : 12.60, Ref FoV %: 7.10, TOL=15.0,
img: 320x240, ref: 2592x1944
01-24 16:03:22.377 ERROR Exception occurred in test_video_aspect_ratio_and_crop.
Traceback (most recent call last):
File "/home/vantron/.local/lib/python3.8/site-packages/mobly/base_test.py", line
783, in exec_one_test
test_method()
File "/media/vantron/E0AE7CF1C8CC5D66/Android14/android-cts-verifier-14_r6-
linux_x86-arm/android-cts-
verifier/CameraITS/tests/scene4/test_video_aspect_ratio_and_crop.py", line 308,
in test_video_aspect_ratio_and_crop
raise AssertionError(f'{e_msg}check failed.')
AssertionError: FoV check failed.
```

6.3.5.3 原因分析

客户反馈是录像分辨率中添加1280x720分辨率后导致上述失败问题。
经分析未加1280x720分辨率前，ITS测试分辨率流程为：
176x144(LOW)->640x480(HIGH)->176x144(QCIF)->640x480(480P)->320x240(QVGA)
加上1280x720后，ITS测试分辨率流程为：
176x144(LOW)->1280x720(HIGH)->176x144(QCIF)->640x480(480P)->1280x720(720P)->320x240(QVGA)

中间有1280x720后：
RKISP1CameraHw::checkNeedReconfig中检测不进行config切换，关键LOG如下：
I RkCamera: <HAL> ImguUnit: @configStreams 179: configChanged :0

导致还是使用1280x720的mainpath配置；从而postpipeline中还是用1280x720去裁剪，从而导致异常，LOG如下：
D RkCamera: <HAL> PostProcessPipeline: processFrame: crop region(880,720,200,0) from (1280,720) to 176x144, infmt 0,842094158, outfmt 34,842094158

如果重新配置，是按照isp这边裁剪计算，就会正常；
D RkCamera: <HAL> GraphConfig: @cal_crop : src_ratio:1.333333, dst_ratio:1.222222, src(1188x972), dst(176x144)

6.3.5.4 解决办法

`sdk/hardware/rockchip/camera$` 目录添加如下修改，如没有更新SDK或找FAE获取

```
commit adb54f0cd64626b0b02086355bd4d76cb3ffa976
Author: Wang Panzhenzhuan <randy.wang@rock-chips.com>
Date:   Tue Mar 18 16:33:58 2025 +0800

    fix ctsv its test_video_aspect_ratio_and_crop fail

    need reconfig stream for different w/h ratio to fix
    test_video_aspect_ratio_and_crop fail

    Test: CTSV ITS
    scene4 test_video_aspect_ratio_and_crop.py

    Type: Fix
    Redmine ID: #526612
    Associated modifications: N/A
    Test: ITS test

    Signed-off-by: Wang Panzhenzhuan <randy.wang@rock-chips.com>
    Change-Id: I655e783892915f7eead1ff11ab3d7f30f05fb998
```